AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 9 · STUDENT LESSON

Features and Distractors

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Choose the target. Preserve the distinction. Challenge the shortcut.

Use controlled cards, explicit timing and diagnostic subject-logo comparisons to justify inputs.

Keep one habit: Task and units → missing distinction → controlled changes → honest evidence limits.

MAIN LESSON · SECTION 1

What information does the task actually need?

> Choose inputs that preserve needed distinctions before considering a model.

Your learning goals

I can choose recorded properties that preserve distinctions needed for a stated task.

I can show why identical **feature record**s can hide examples with different **target label**s.

I can reject an answer copied into the **input**s and test a proposed **feature** with a new case.

Retrieve last week's label-evidence limit

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Two annotators agree that a calendar card is ready. What must you still check before treating that label as correct?

This week's controlled-test goal

By the end, I can test a distracting **feature** with controlled changes rather than confounding clues.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

After trying, check the **relevant** lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 9: **Feature**s and **Distractor**s

A **feature** should help the task and be available at decision time.



THINKER MISSION

Rank book **feature**s; reject cover color as a **shortcut**.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Define the card-fitting task before choosing features

> The rule uses measured dimensions in a fixed orientation.

The exact task and measurement assumptions

Imagine a paper sorting task for a model delivery station. Each rectangular card has a length, width, colour and card number.

The station labels a card fits if its length is at most 12 centimetres and its width is at most 8 centimetres. Both limits include equality.

Cards stay flat in the marked orientation; turning, bending and thickness are outside this paper task. All measurements in the first examples are known positive whole numbers.

Separate the feature record from the target

A **feature** is a recorded **input** property, such as width in centimetres. The **target label** is the answer we want to produce: fits or too large under this guide. Decide what the answer means before choosing the **input**s.

A card number can help us find a record, but knowing its number does not tell us its dimensions.

Why this investigation needs no trained model

This guide already gives us a direct rule, so we do not need to train a model to solve it. We use the task to investigate the information that either a rule or a **learned model** would need.

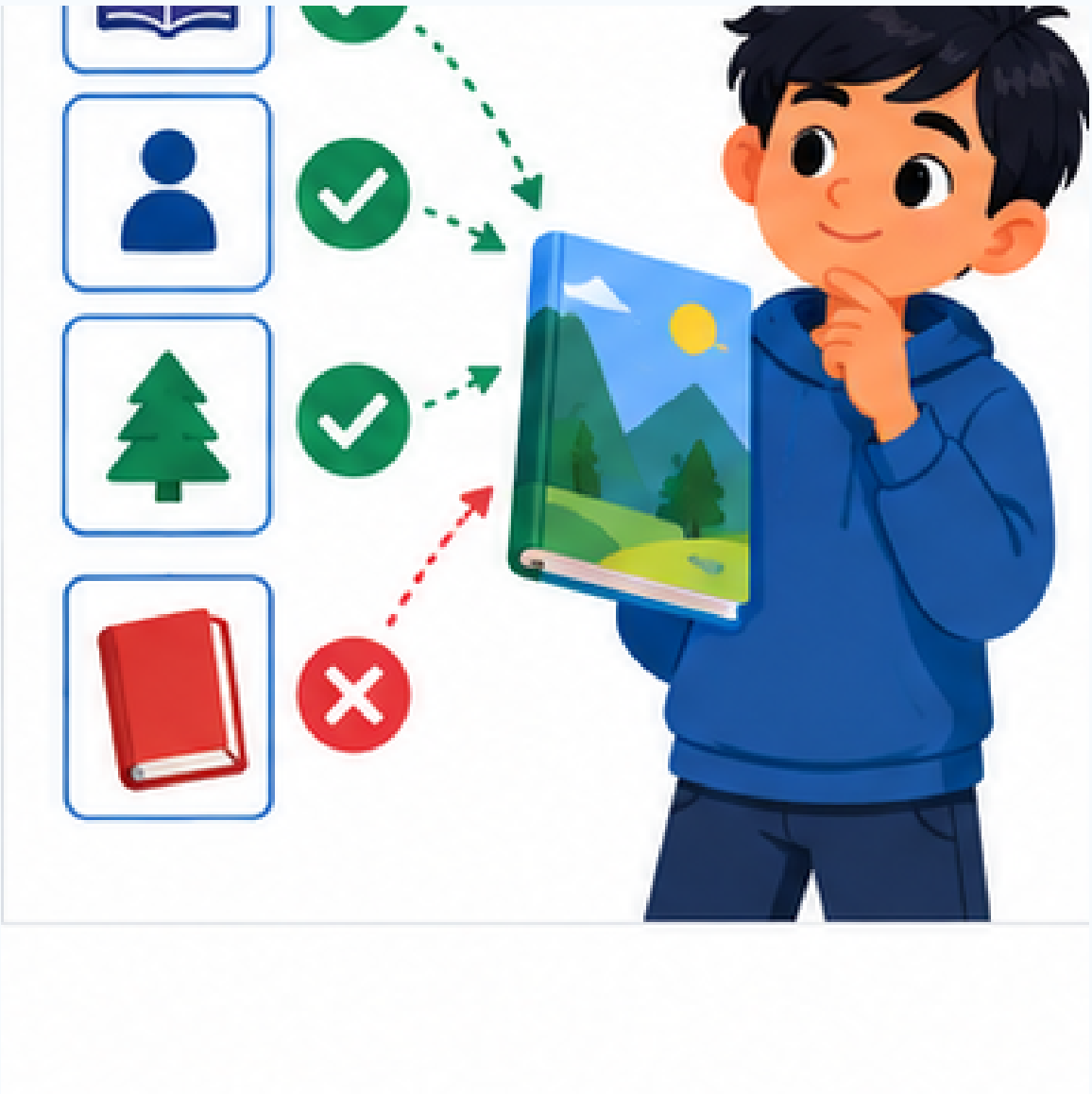
Later, when relationships are learned from examples, the same question remains: did we supply the distinctions the task requires?

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **feature**s and **distractor**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

See what disappears when width is omitted

> The same visible input cannot guarantee two different correct answers.

Compare cards A and B

Card A is 10 centimetres long and 7 centimetres wide. Card B is 10 centimetres long and 9 centimetres wide. A fits; B is too large because its width exceeds 8. Compare two ways of recording these cards.

Follow the length-only and length-width records

AI ACADEMY / AI-2/9 · 03 · INSPECT THE LESSON MODEL

The same cards under two feature choices

Original card	Choose the inputs	Resulting feature record
A is 10 by 7 cm and fits	Keep length only	10 cm
B is 10 by 9 cm and is too large	Keep length only	10 cm
A is 10 by 7 cm and fits	Keep length and width	10 cm and 7 cm
B is 10 by 9 cm and is too large	Keep length and width	10 cm and 9 cm

Length alone makes A and B look identical to the system. Retaining width preserves the difference needed by this task. The letters identify the examples here; they are not additional inputs.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the information limit

A system receiving only the length cannot use the hidden width to tell these two cards apart. The same **input**, 10, is paired with two different correct answers.

It may guess, return an uncertainty, or use a usual answer, but length alone cannot guarantee the right label for both. A more complicated model cannot reconstruct this missing distinction with certainty from that **input** alone.

Keep order and units explicit

A **feature record** needs a consistent order and units. Here we write length in centimetres first, then width in centimetres.

The record 10, 7 means something different if someone silently switches the order or records millimetres. Write the definitions alongside the data.

Point to the distinguishing measurement

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

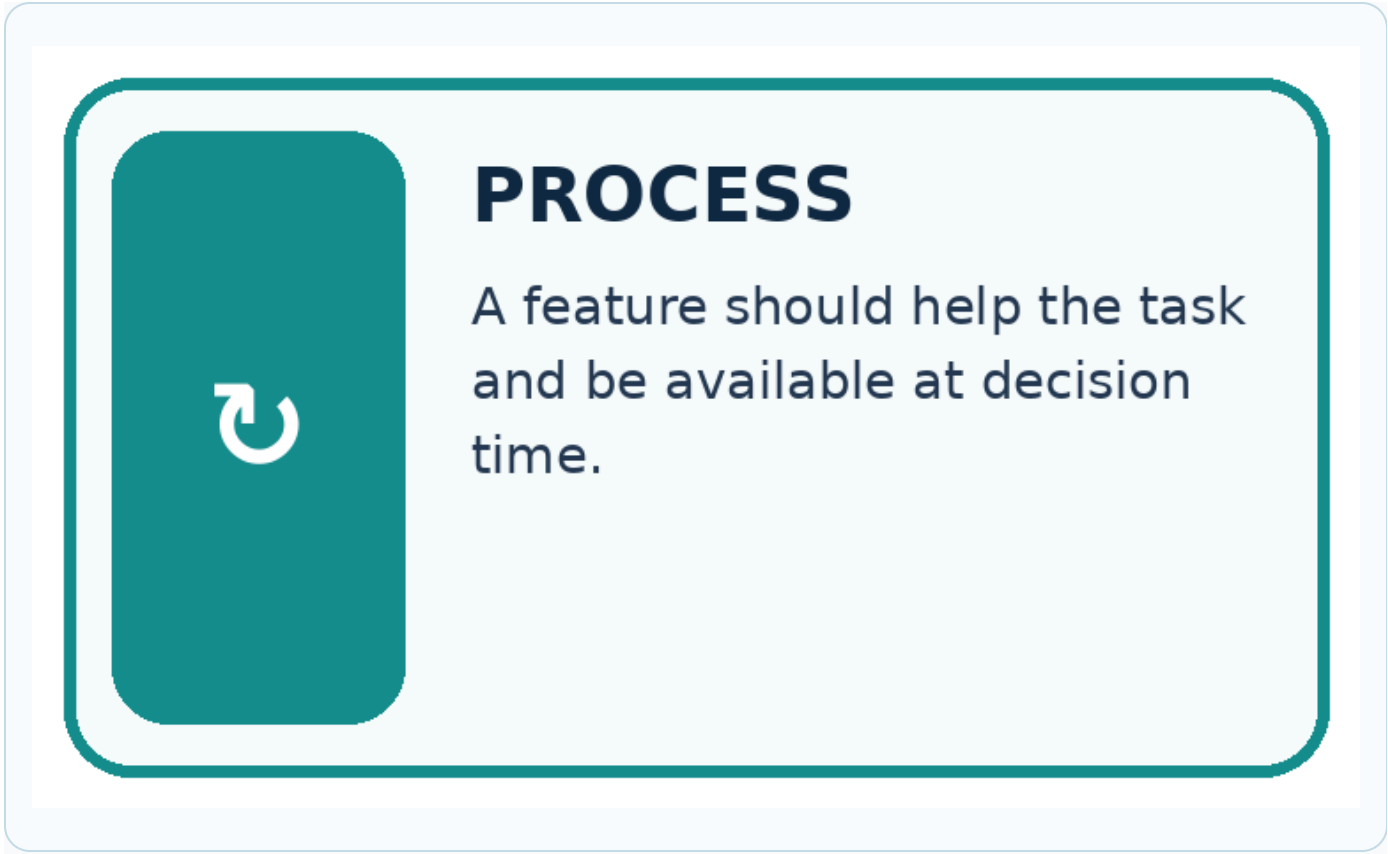
VISUAL **MECHANISM**

Input → **Process** → **Output**

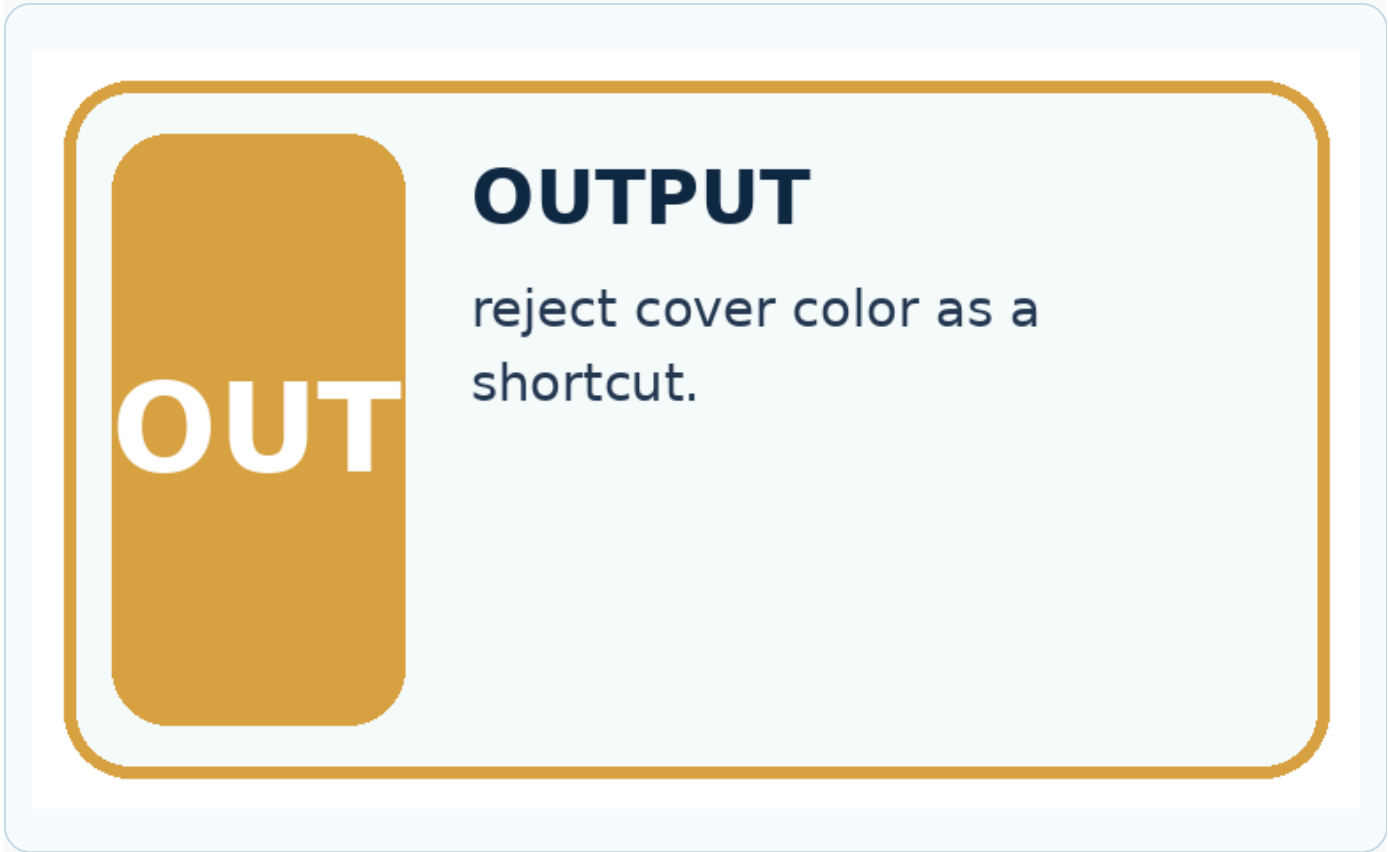
Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
A feature should help the task and be available at decision time.

↓


OUT
OUTPUT
reject cover color as a shortcut.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

The same cards under two feature choices

ORIGINAL CARD	CHOOSE THE INPUTS	RESULTING FEATURE RECORD
A is 10 by 7 cm and fits	Keep length only	10 cm
B is 10 by 9 cm and is too large	Keep length only	10 cm
A is 10 by 7 cm and fits	Keep length and width	10 cm and 7 cm
B is 10 by 9 cm and is too large	Keep length and width	10 cm and 9 cm

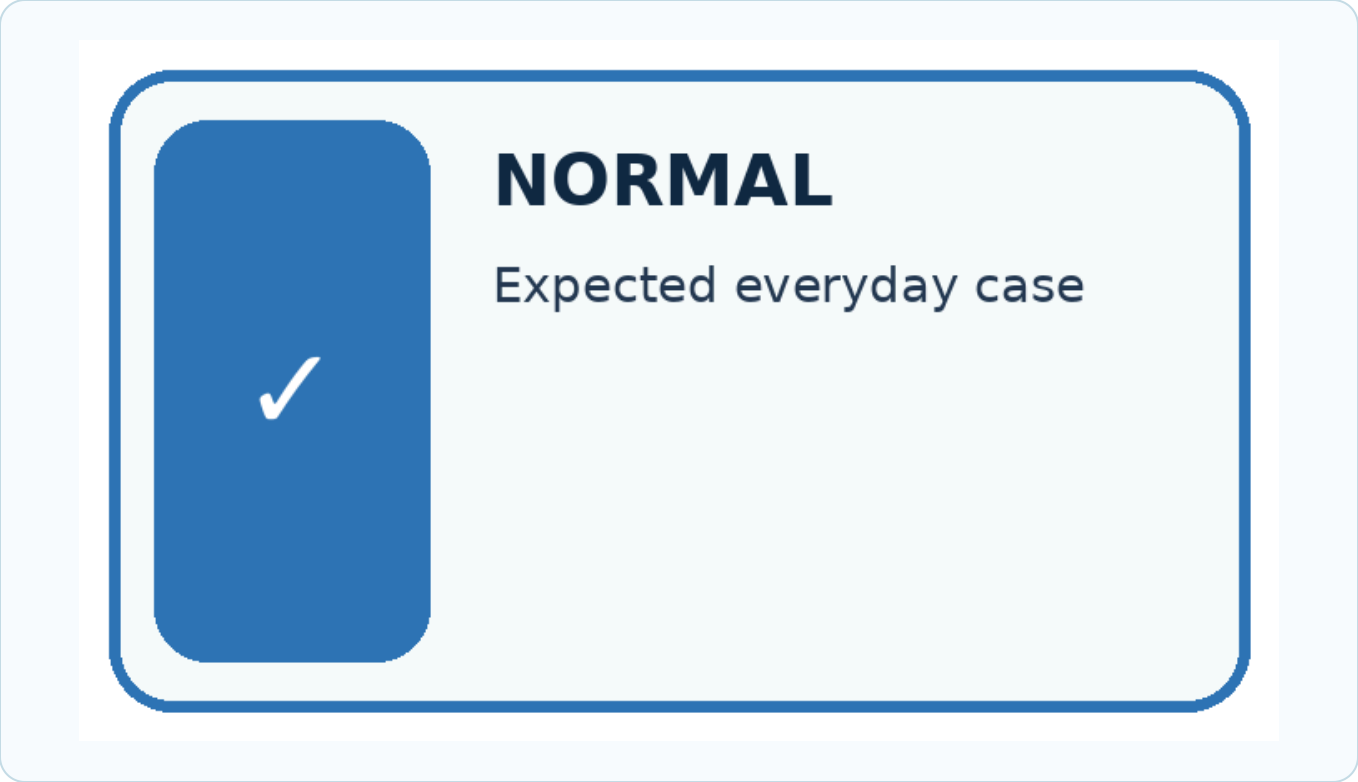
Length alone makes A and B look identical to the system. Retaining width preserves the difference needed by this task. The letters identify the examples here; they are not additional inputs.

Connect the picture and the explanation

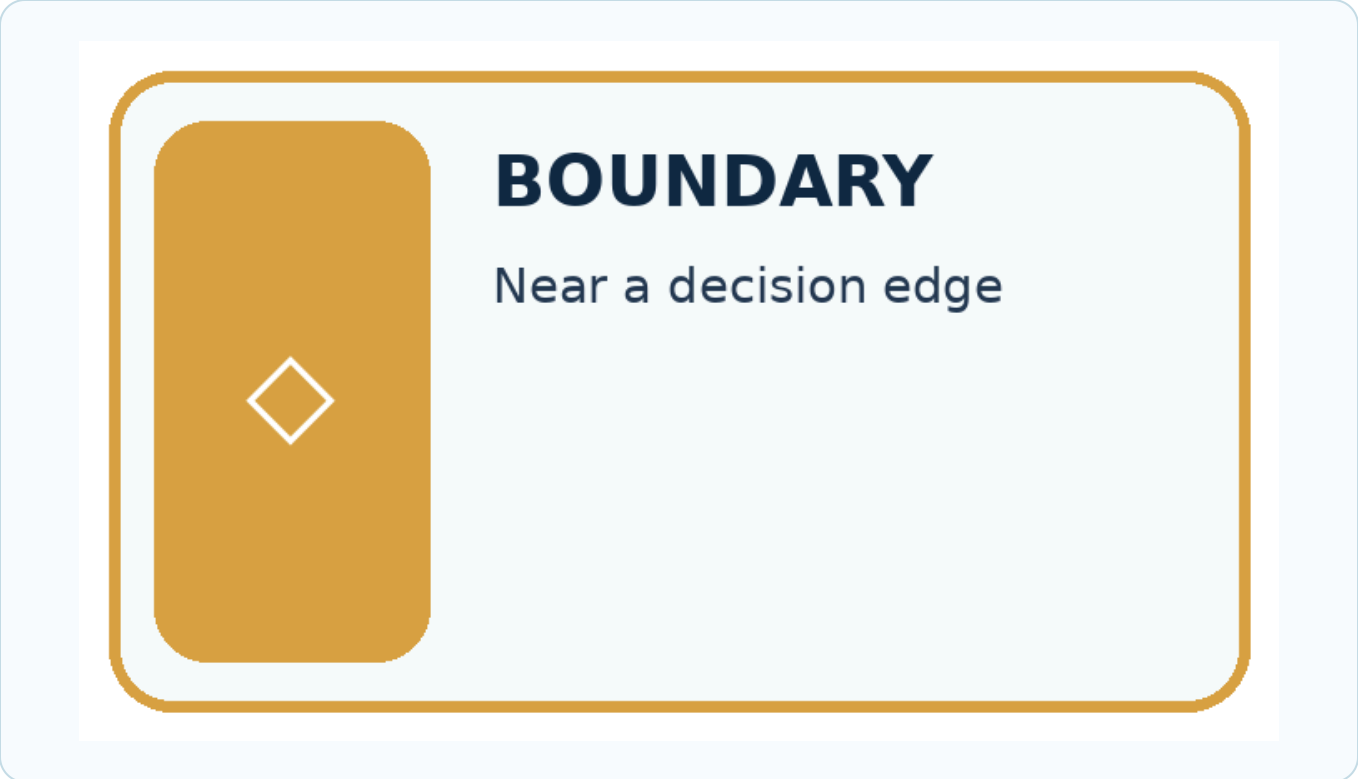
Choose a case that challenges a feature shortcut

> State what is changed and what remains fixed.

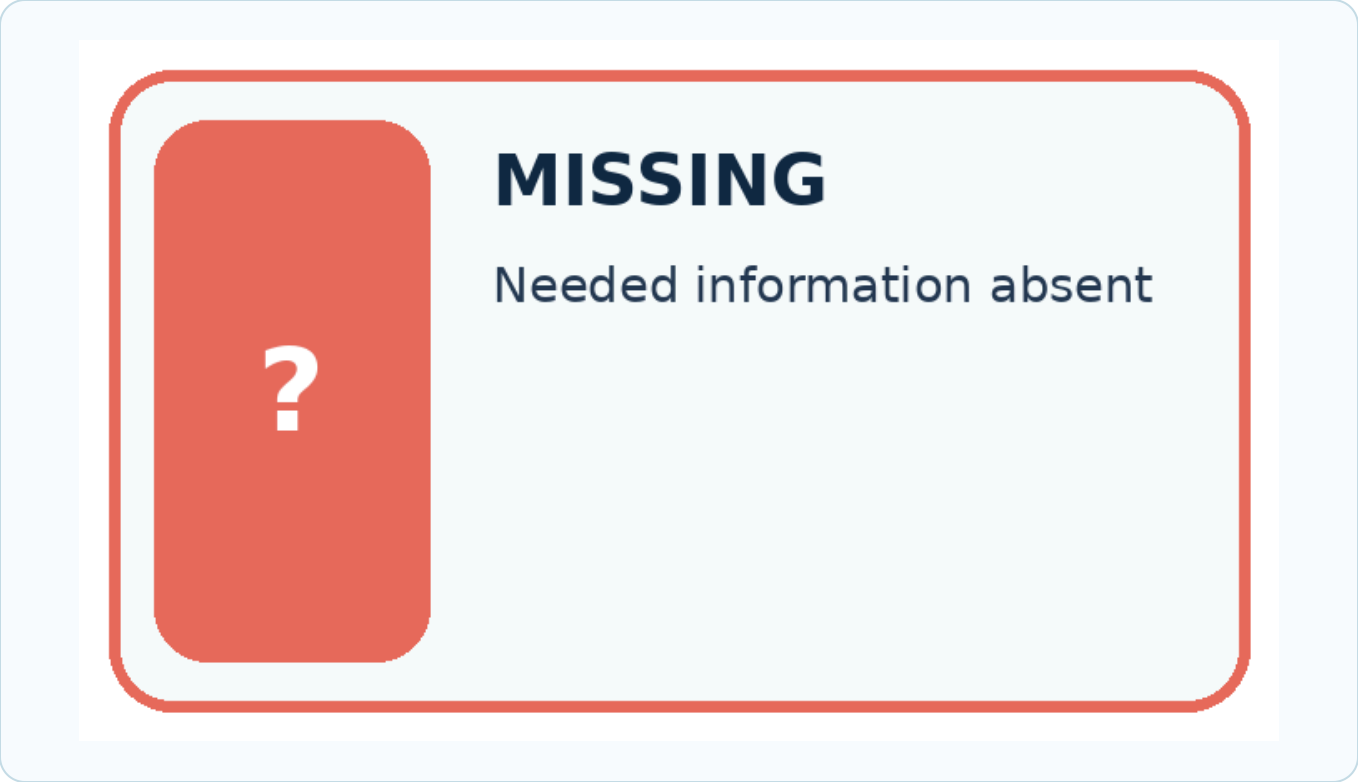
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: The first controlled pair yields mystery for triangle and science for circle.

▼ Check after predicting

Prediction: That pair is **evidence** of unwanted logo sensitivity.

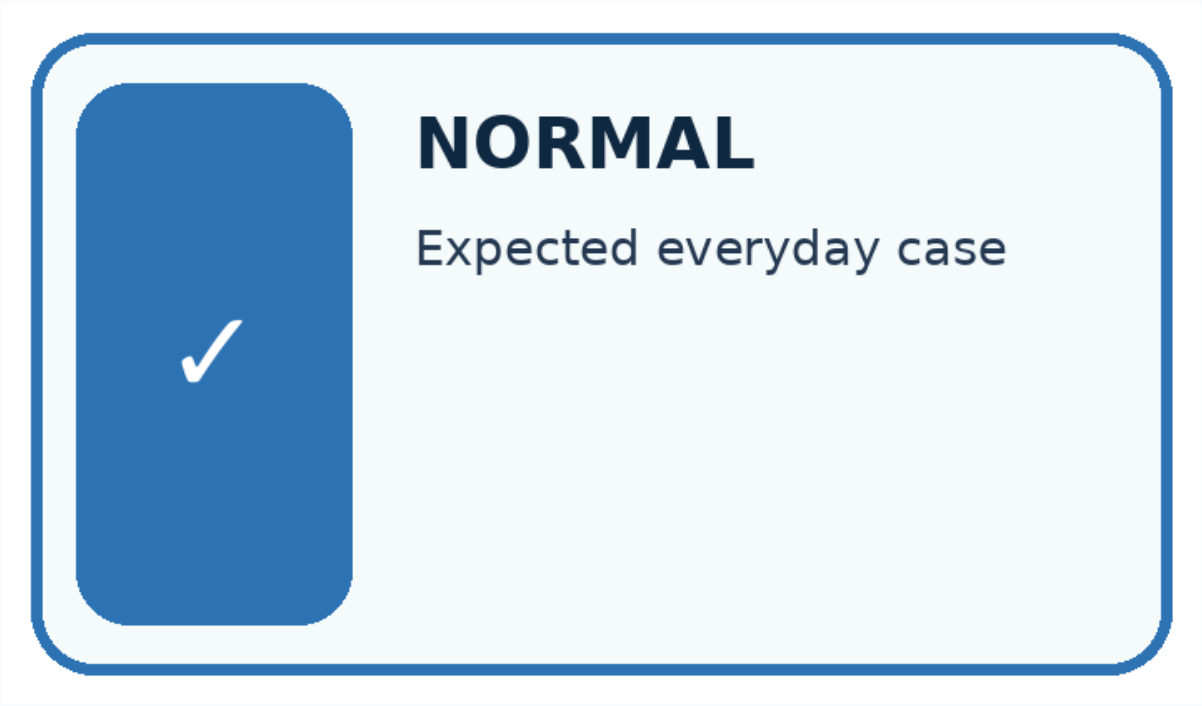
Reason: The subject description stayed mystery; only the ir**relevant** logo changed.

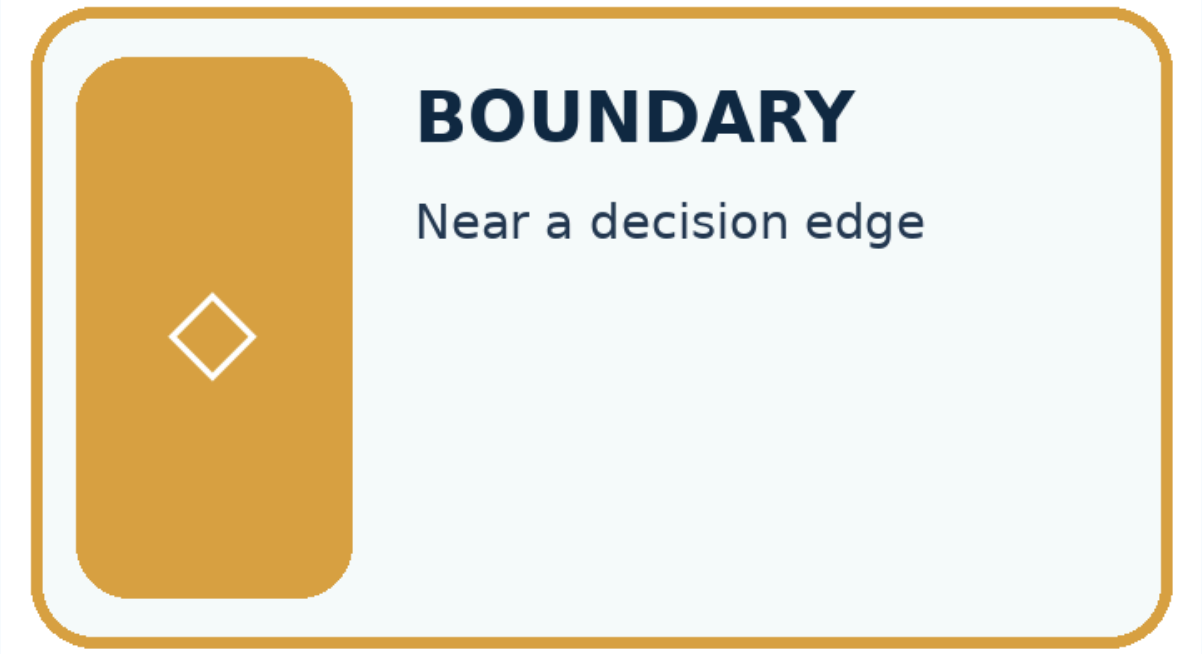
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Compare controlled changes, then investigate a topic-logo shortcut

> Change one aspect at a time and label predictions honestly.

Separate rule-derived answers from unobserved model outputs

Keep two investigations separate. In the card task, the written dimension rule determines the expected answers. In the fictional classifier case below, the old training combinations are given but new test predictions are not. You can design diagnostic pairs without pretending to have run a model. The changed-case panel supplies hypothetical outcomes for interpretation only.

Four card comparisons with one changed input

Card measurements and colour	Label under the guide	What the comparison shows
10 by 7 cm and blue	Fits	Compare with the next card
10 by 9 cm and blue	Too large	Same length and colour but width changes the label
13 by 7 cm and blue	Too large	Same width and colour as the first card but length changes the label
10 by 7 cm and red	Fits	Changing only colour does not change this guide

Explain useful dimensions and a misleading colour pattern

The first two cards show why width matters. The first and third show why length matters. The first and fourth show that colour has no part in this particular rule.

That is a statement about our task, not a claim that colour is useless for every task. For sorting red and blue cards, colour would be essential.

Suppose every **fitting** card in a small practice set is blue and every oversized card is red. Colour would appear to predict the labels perfectly in that set. A red 10 by 7 card and a blue 13 by 7 card expose the **shortcut**.

Check a proposed **feature** on deliberately different cases before trusting the practice score.

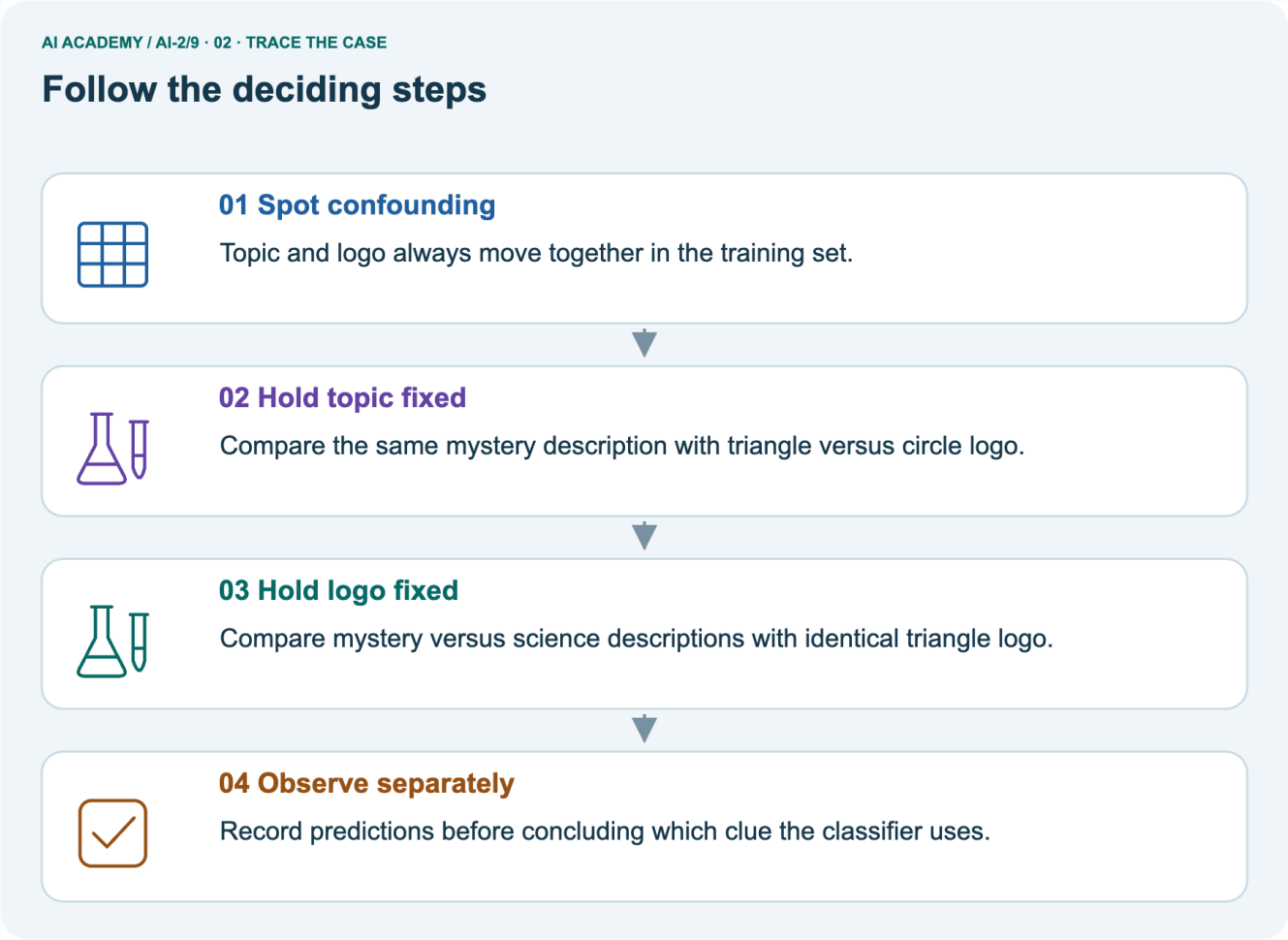
Introduce the separate topic-classifier case

A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos.

Training contains no mystery-circle or science-triangle covers.

Goal is to identify topic from actual subject information, not publisher logo. New controlled test **output**s have not been supplied.

Design pairs that separate subject from logo



Follow the labelled picture with the matching explanation. Open the vector diagram

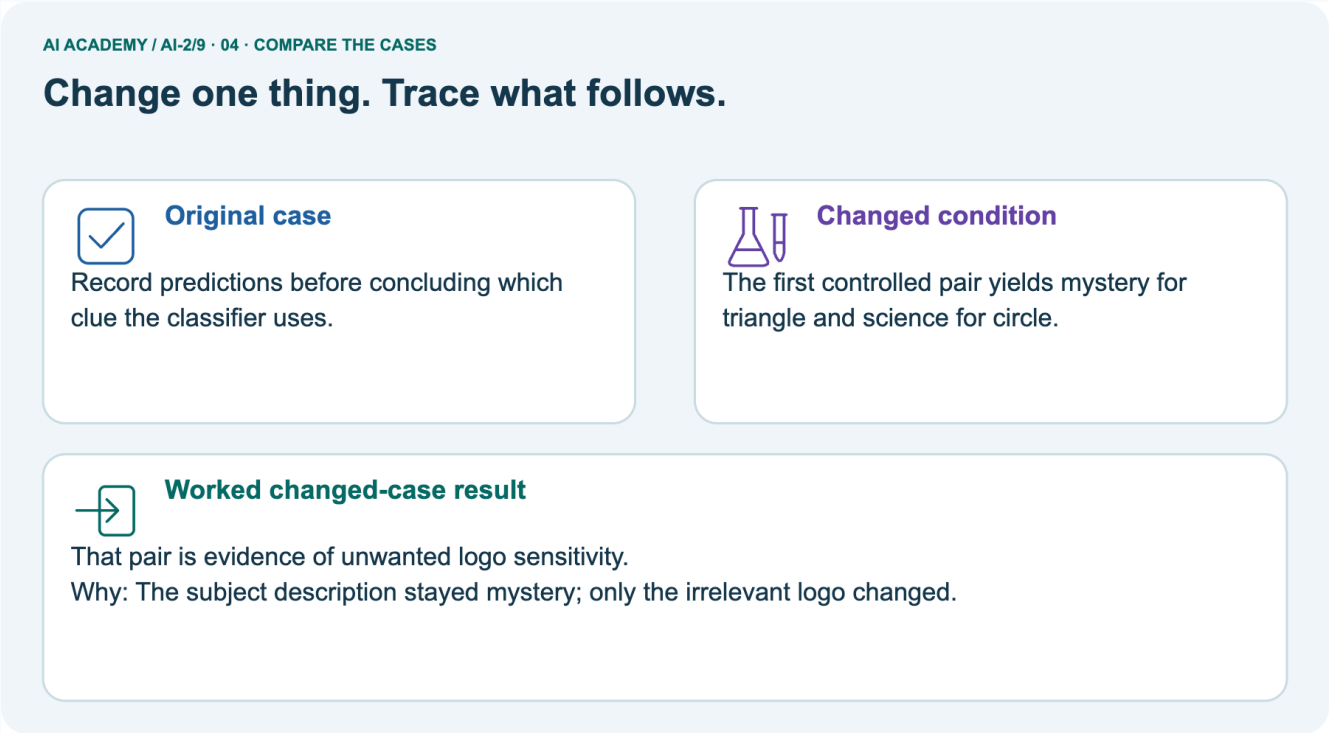
Either subject information or logo could explain success on the old combinations. Controlled pairs separate their **prediction**s by changing one aspect at a time.

Designing the pair is not equivalent to observing the model's response. Keep the descriptions identical when testing logo so differences do not have two possible causes.

Interpret the supplied hypothetical changed case

- Change one thing: The first controlled pair yields mystery for triangle and science for circle. Predict the result before checking the explanation. What remains the same?

▼ Check that interpretation after trying



Follow the labelled picture with the matching explanation. Open the vector diagram

Limit the conclusion to the evidence

- Read the labelled picture: Does this one pair prove the classifier always ignores subject? Explain a narrower claim.

- Controlled **observation**s support specific sensitivity claims; they do not reveal every internal **feature** a **learned model** uses.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 9 • WORKED EVIDENCE

1. Goal: recommend enjoyable readable books.
2. Useful: genre match, reading level, length preference.
3. Distractor: cover color may match past labels by accident.
4. Deployment check: every chosen feature must exist before recommending.

1. Goal: recommend enjoyable readable books.

2. Useful: genre match, reading level, length preference.

3. **Distractor**: cover color may match past labels by accident.

4. Deployment check: every chosen **feature** must exist before recommending.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

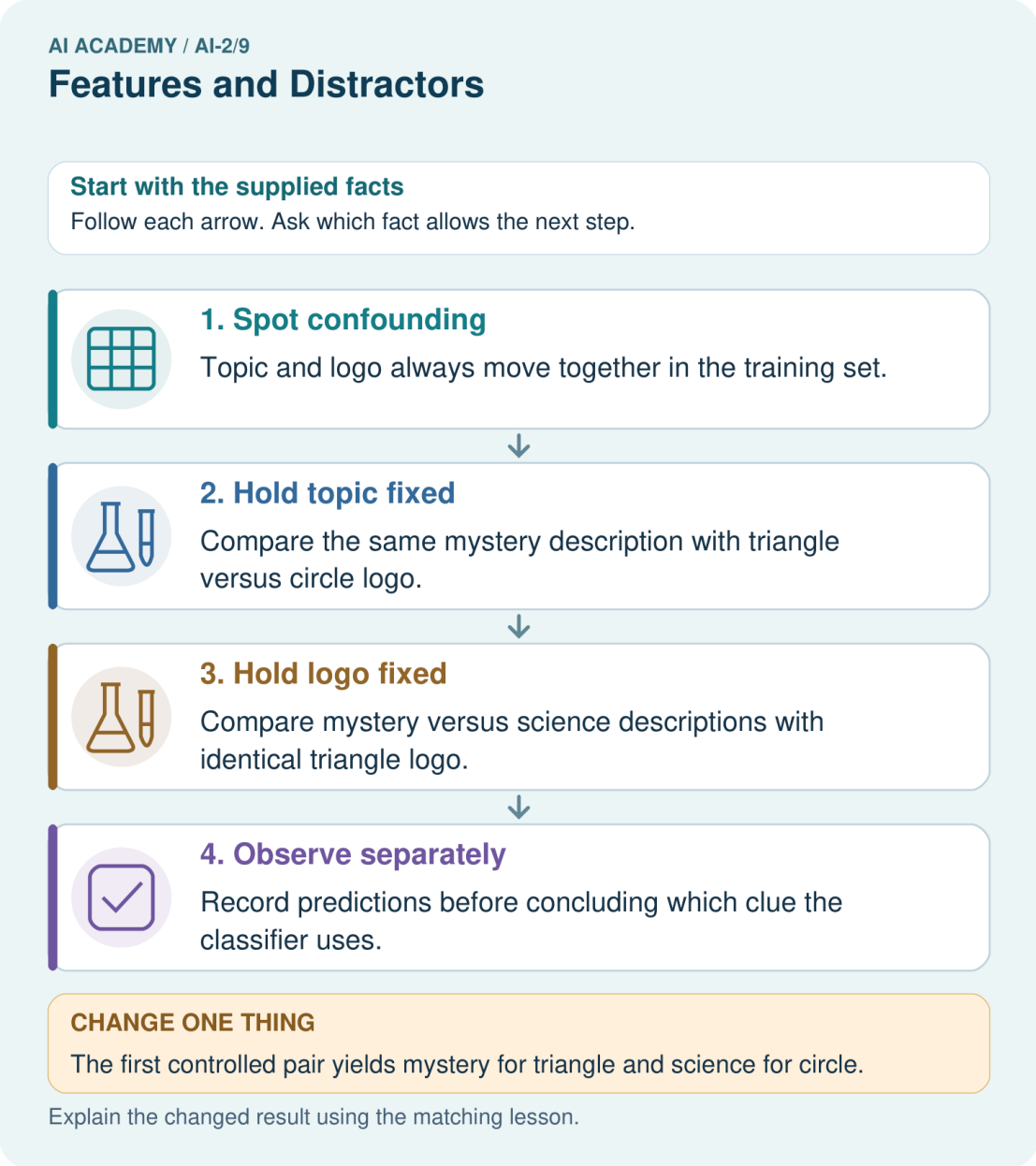
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Features and Distractors

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Spot confounding	Topic and logo always move together in the training set.
2. Hold topic fixed	Compare the same mystery description with triangle versus circle logo.
3. Hold logo fixed	Compare mystery versus science descriptions with identical triangle logo.
4. Observe separately	Record predictions before concluding which clue the classifier uses.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Challenge a perfect practice score with a diagnostic card

> A shortcut can match a small practice set without following the task.

Check distinguishability and input timing

Before adopting a **feature** set, state the task, name the **input**s with units, and ask whether two different answers can share the same **input** record. Construct such a pair if you can.

Then check that each **input** is available at the time the answer is needed. These checks can reveal a broken representation before any model is trained.

Retain the paper task's limits

For our known measurements and fixed orientation, length and width are sufficient to apply the paper rule exactly.

That does not establish whether a real parcel will fit through a real opening: thickness, shape and measurement error were excluded. If the task changes, review the **feature**s again.

More columns are not automatically better; each one needs a reason to be collected and used.

Create a counterexample to colour-only prediction

> **Someone claims that a model using only card colour is reliable because it matches every practice label. Give a new card that challenges this claim and explain what should be checked instead**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Any field correlated with past choices is automatically safe to use.
Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Put every candidate input on the decision timeline

> An answer added after inspection cannot be an earlier prediction input.

Spot the sticker leakage

After measuring a card, a teacher adds a sticker saying fits or too large. A learner proposes using sticker text as an **input** to predict the label before that inspection.

The sticker seems excellent in the completed records because it contains the answer. But it does not exist at the time of the intended **prediction**. This is **leakage**: development has used information the real task cannot provide.

Draw a line at the moment the system must answer. On the before side, put the permitted measurements available then. On the after side, put the final label sticker.

Keep labels available for teaching and checking **prediction**s, but do not feed the sticker back in as an **input** for this task.

Try an unknown width before inspection

> **A card is 12 cm long and its width is unreadable. Someone proposes using the label sticker that will be added after inspection. Which information is usable before inspection? Explain why it does not establish the label**

Construct a width-only collision in W2

Use workbook W2 for the connected example “**Feature**s and useful distinctions”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge the missing-measurement guarantee

- Use the worked case for **Feature**s and useful distinctions. Proposed explanation: “A stronger model can always recover an omitted measurement”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain what a controlled pair can establish

Does this one pair prove the classifier always ignores subject? Explain a narrower claim.

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.

TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Practise with less help: Prove that one measurement is insufficient

Check your reasoning

Explain the picture

My explanation using a pictured step and the given facts

MAIN LESSON · SECTION 9

Audit eight supplied candidate features

> Select the smallest justified set for this explicit task.

Fix the task and decision time

Feature audit: predict fits or too large for positive whole-number rectangular cards, flat in the fixed orientation. Fits means length <=12 cm AND width <=8 cm, including equality. Make the prediction before inspection. Review these eight candidate fields; an available field is not automatically useful. There is no requirement to select three when two are sufficient for this written rule. Use fictional records and collect no real learner names.

The eight candidates

Candidate field	Meaning and timing
length_cm	Measured before inspection
width_cm	Measured before inspection
colour	Written red or blue before inspection
card_number	Identifier assigned before inspection
thickness_mm	Measured before inspection; excluded by the flat-card task
inspection_sticker	Fits/too large answer added after inspection
learner_full_name	Name attached to a record before inspection
record_time	Time when the record was written

Justify, challenge and record

1. For each field record: available before the decision? needed by the stated rule? unnecessary information about a person? possible shortcut or leakage? Justify your selected set and reject at least two tempting alternatives. Then use 10×7 and 10×9 cm to challenge length-only inputs, and 11×6 and 13×6 cm to challenge width-only inputs. Keep all cards blue for these pairs. Finally compare blue and red copies of the same 10×7 card. Write the rule-derived expected labels; record paper checks separately from model predictions, which remain not supplied.

Keep expected and actual evidence separate

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

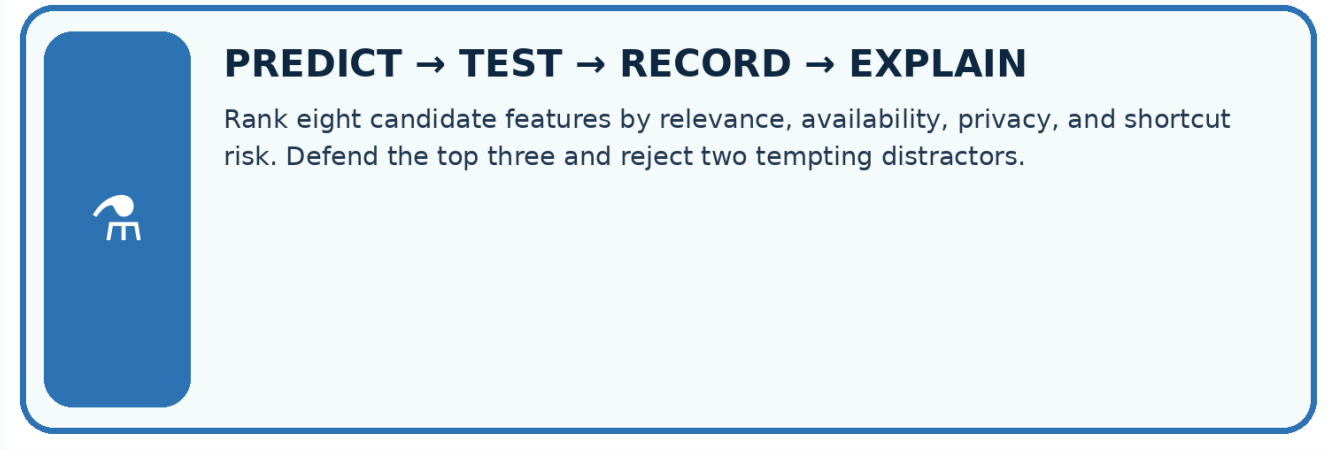
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Audit the eight supplied candidate **feature**s for relevance, availability, privacy and **shortcut** risk. Defend the smallest sufficient set for the written card rule and reject at least two tempting alternatives; do not add a third **feature** merely to meet a quota.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: test subject versus background colour

> Design controlled comparisons without inventing classifier outputs.

The reserved space-and-animal case

Training covers show space books only on yellow backgrounds and animal books only on blue backgrounds. Design two controlled comparisons to test whether a classifier uses the intended subject.

Your first attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Week 9 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Predict whether a book is short using page count and cover colour. Which **feature** directly supports the defined label? Give a counterexample to colour.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one **relevant** detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Use explainable traits that connect to the reader's stated preferences.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

All animal books in **training** have green covers. How could you test whether a recommender is using subject or cover colour? Supply two diagnostic test descriptions.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Justify every recommender feature against the stated task

> Keep subject information separate from catalogue styling.

Your portfolio checkpoint

THIS WEEK'S CHECKPOINT Use explainable traits that connect to the reader's stated preferences.

Test recommender **feature**s for **shortcut**s, keeping **relevant** subject information separate from superficial catalogue styling.

Record inputs, rules, human choice and limits

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

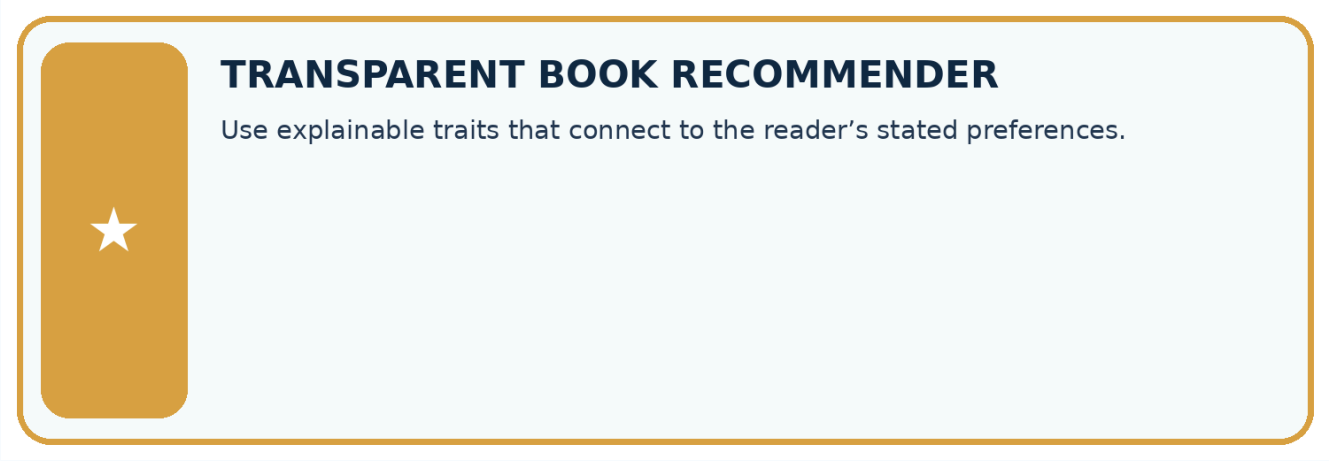
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain a distinction the inputs must preserve

> State a diagnostic pair and a limit of your evidence.

Recall the goal without notes

- Close your notes. Explain this goal in your own words: Choose recorded properties that preserve distinctions needed for a stated task

My deciding step, evidence and one remaining question

Choose a changed case for your return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
feature	An input attribute or representation used by a model or decision rule.	
relevant	Useful to the particular question, target or decision being considered.	
distractor	A detail that draws attention but does not help answer the current question.	
shortcut	A convenient but unreliable cue used instead of the relationship the task requires.	

Terms for the connected explanation

Word	Meaning
Target label	The answer a supervised learning task asks a model to predict
Feature record	The selected input values for one example, in a defined order

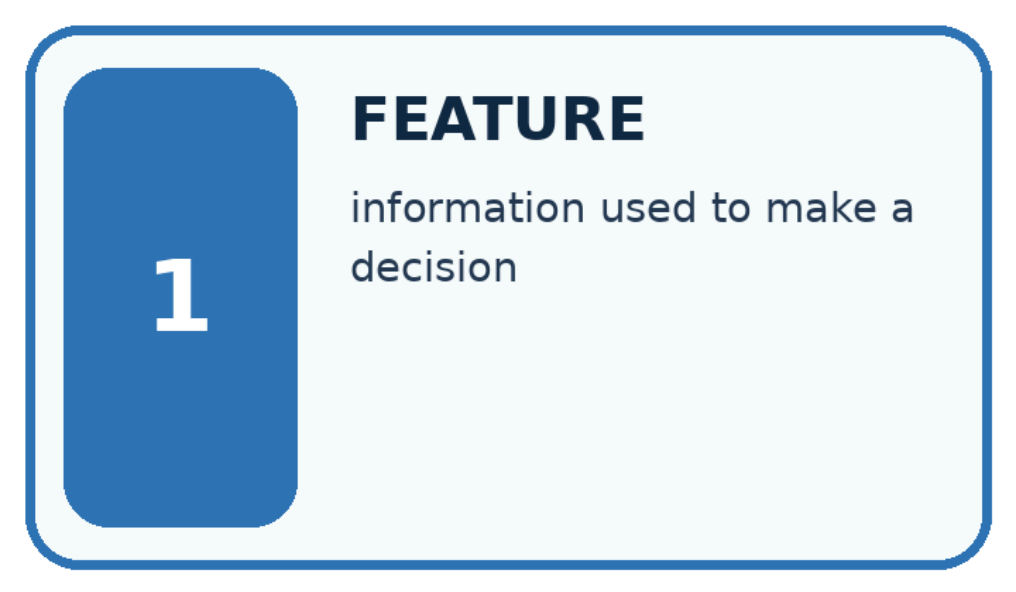
Word	Meaning
Relevant feature	An input property that helps make distinctions needed for the stated task
Leakage	Information used during development that would not legitimately be available when making the intended prediction

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

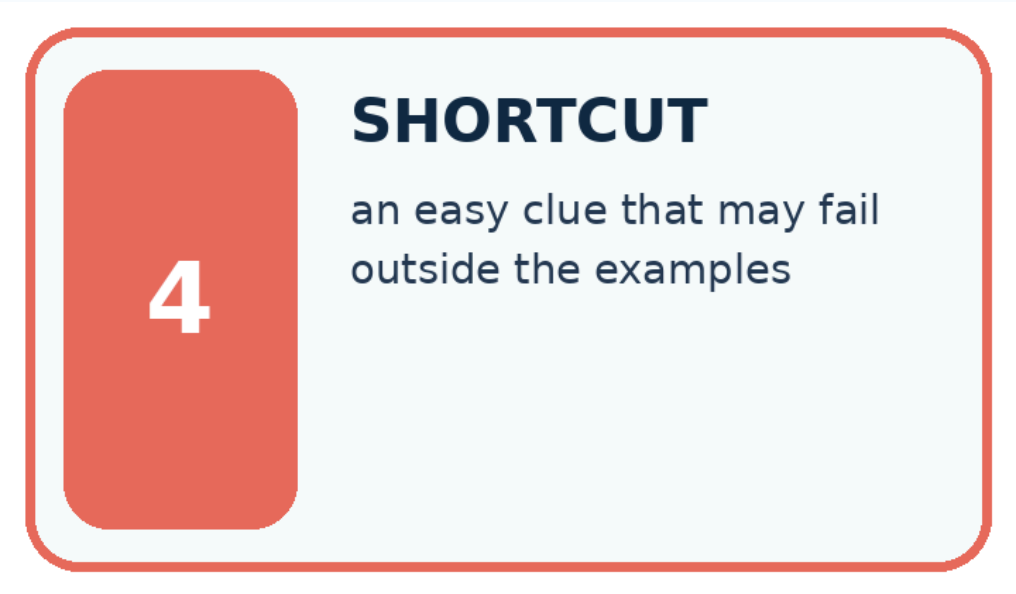
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
RELEVANT
connected to the goal


3
DISTRACTOR
available but unhelpful information



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

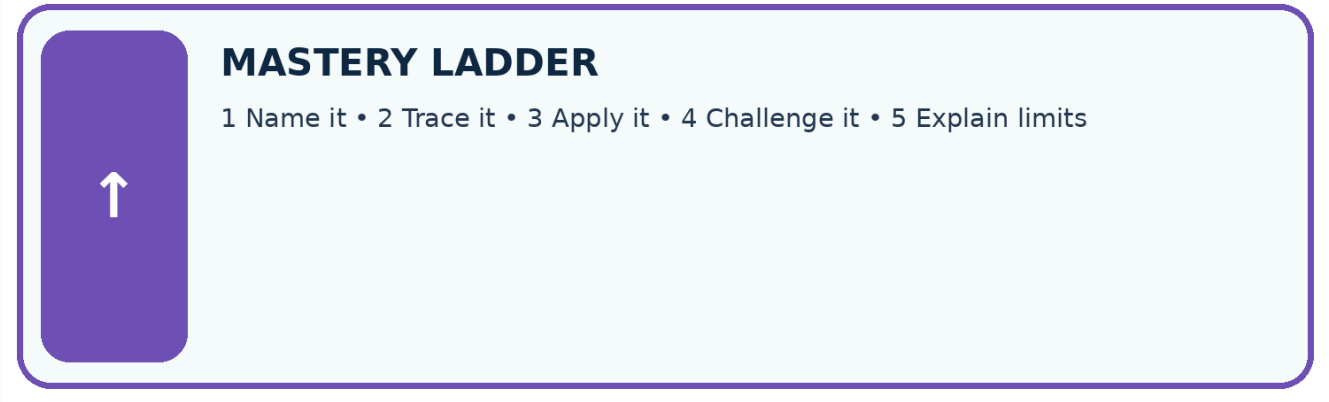
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Changed independent case is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

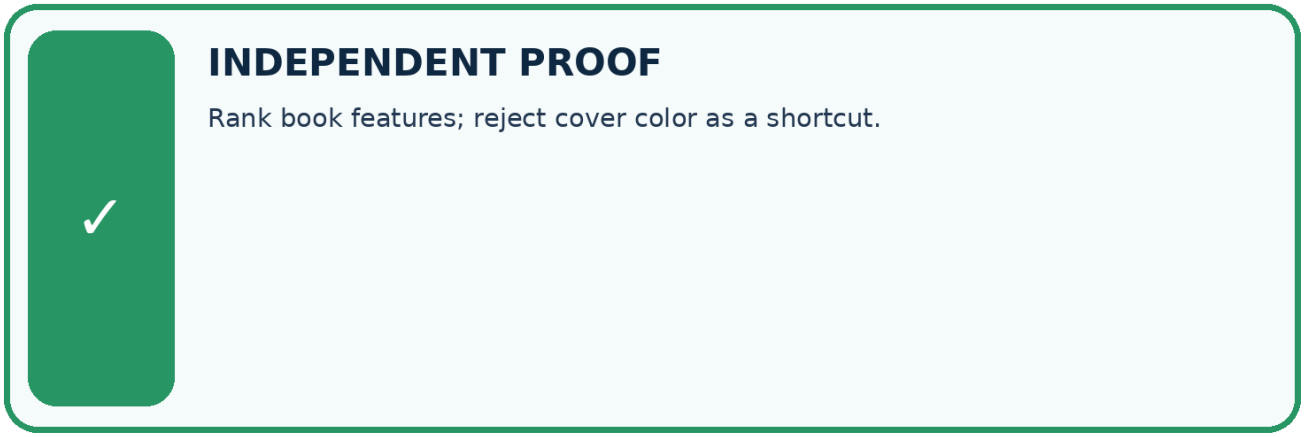
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Changed independent case is the selected independent case.



EXPLANATION CHECK Preserve required distinctions, test one-factor changes and separate expected labels from actual model observations.

Write or draw your reasoning and evidence.

▼ **Extra practice: Useful evidence versus a shortcut**

REFERENCE · SECTION 13

Extra practice: Useful evidence versus a shortcut

Try the question before opening its worked explanation.

Read the supplied case and question

Target: includes animal characters. Fields: animal tag, cover colour and shelf number.

Choose useful **evidence** and possible distractions.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

An accurate animal tag directly addresses the target. Colour and shelf can produce accidental **shortcut**s. Test on new books.

Evidence or stage	Value or result
Target	Contains animal characters
Animal tag	Relevant if accurate
Cover colour	Possible distraction
Shelf number	Possible accidental shortcut

▼ Earlier wording, alternatives and picture transcripts

Week 9 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Useful evidence versus a shortcut

Target: includes animal characters. Fields: animal tag, cover colour and shelf number.

Choose useful **evidence** and possible distractions.

An accurate animal tag directly addresses the target. Colour and shelf can produce accidental **shortcut**s. Test on new books.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Choose recorded properties that preserve distinctions needed for a stated task	Explain the deciding step, show a trace or test, and state one limit.
Show why identical feature records can hide examples with different target labels	Explain the deciding step, show a trace or test, and state one limit.
Reject an answer copied into the inputs and test a proposed feature with a new case	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos.
- Which labelled step matches this detail? Record **prediction**s before concluding which clue the classifier uses.
- 1. Hold topic fixed
- 2. Hold logo fixed
- 3. Observe separately
- 4. Spot confounding
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Observe separately Record **prediction**s before concluding which clue the classifier uses.

Either subject information or logo could explain success on the old combinations. Controlled pairs separate their **prediction**s by changing one aspect at a time.

Designing the pair is not equivalent to observing the model's response. Keep the descriptions identical when testing logo so differences do not have two possible causes.

Changed-case question

- Changed-case practice: The first controlled pair yields mystery for triangle and science for circle. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. That pair is **evidence** of unwanted logo sensitivity.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: That pair is **evidence** of unwanted logo sensitivity. The subject description stayed mystery; only the ir**relevant** logo changed.

Repair and save your work

Review Operational Labels (ai-2/8). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Why can we not resolve the disagreement merely by taking a majority of reviewers?.

Return to this week and explain what you changed.

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?.

Return to this week and explain what you changed.

Review Branching Logic (ai-2/3). Goal: Trace the selected branch in an if elif else chain. Short practice: Give expected labels for –1,0,10,11 and identify the equality boundary for Short..

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

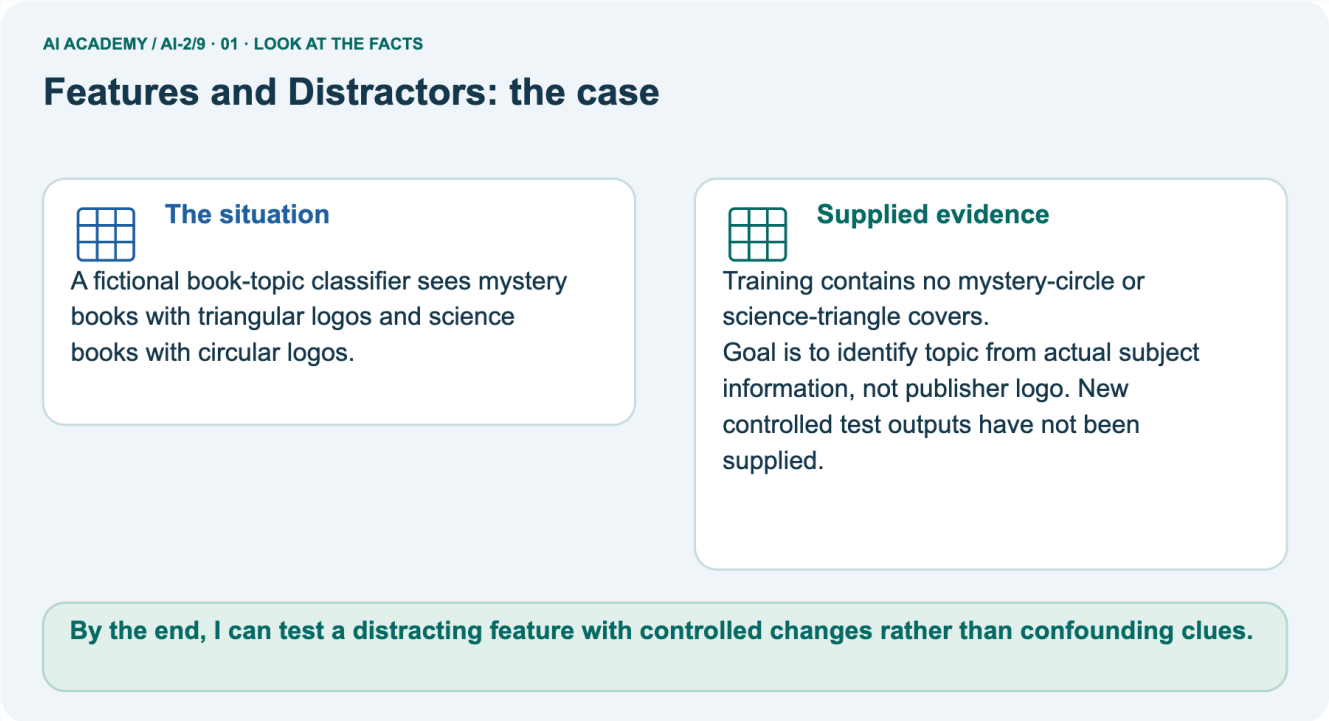
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Features and Distractors: the case



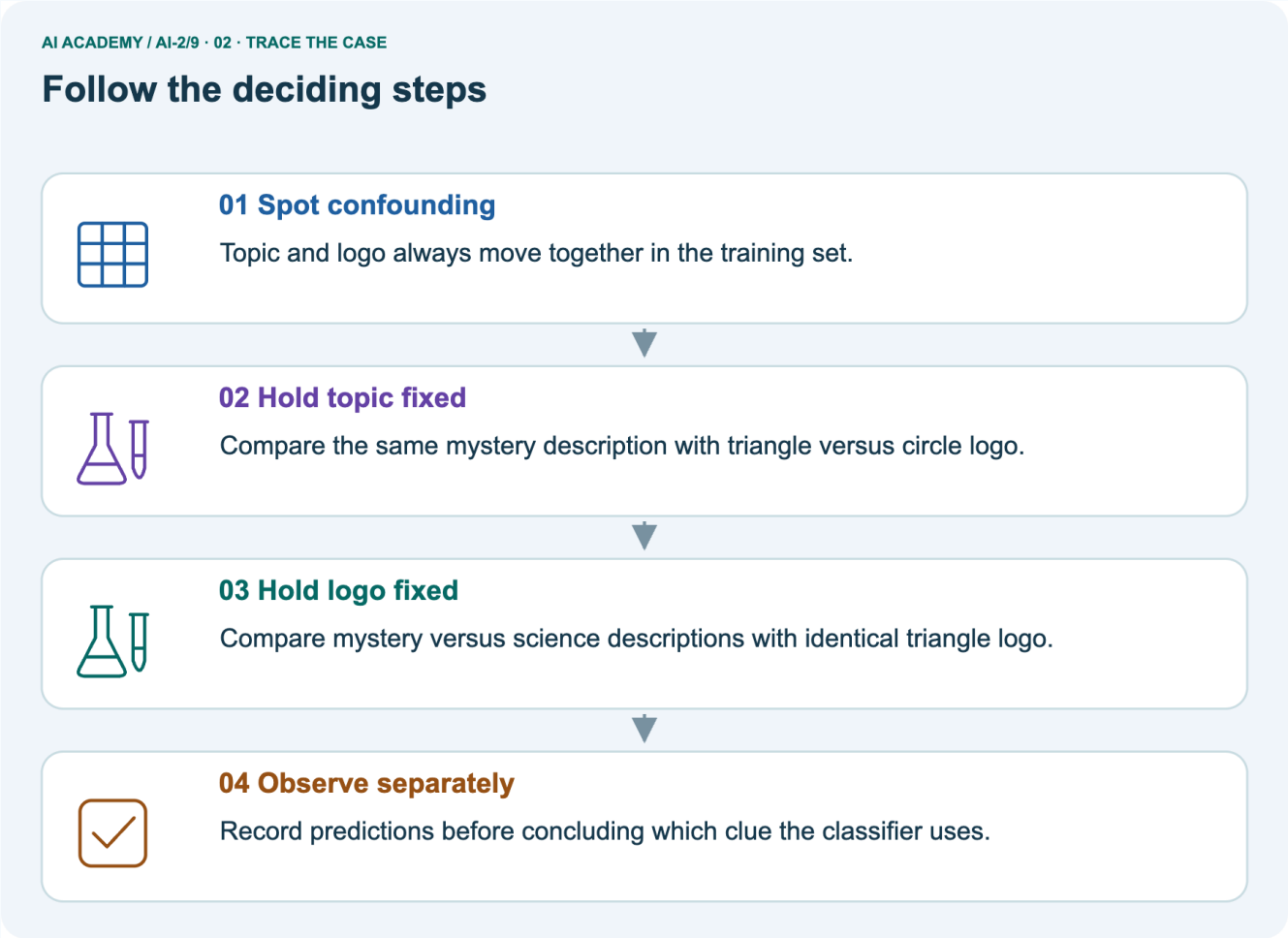
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos. Supplied facts: **Training** contains no mystery-circle or science-triangle covers.

Goal is to identify topic from actual subject information, not publisher logo. New controlled test **output**s have not been supplied.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

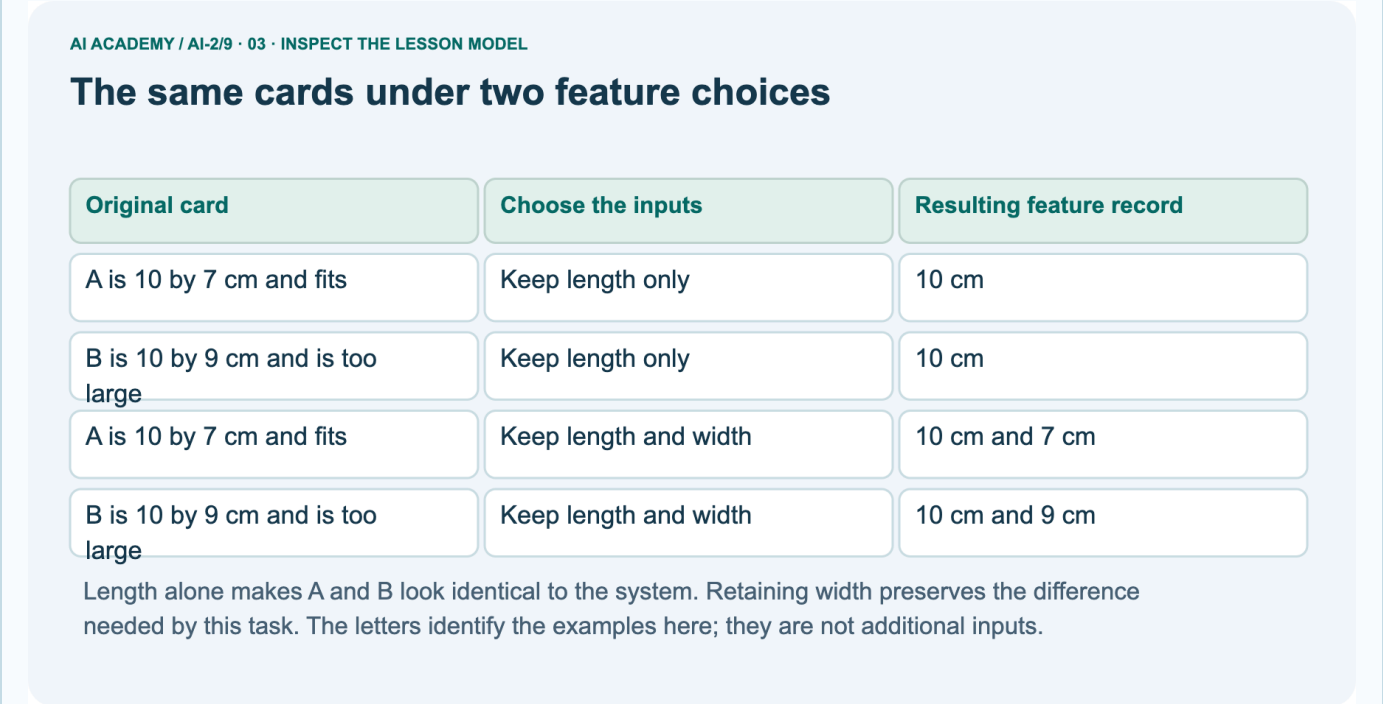


Follow the labelled picture with the matching explanation. Open the vector diagram

Spot confounding: Topic and logo always move together in the **training** set. → Hold topic fixed: Compare the same mystery description with triangle versus circle logo. → Hold logo fixed: Compare mystery versus science descriptions with identical triangle logo. → Observe separately: Record **prediction**s before concluding which clue the classifier uses.

Point to each numbered stage and explain which recorded **input** it uses.

03. The same cards under two feature choices



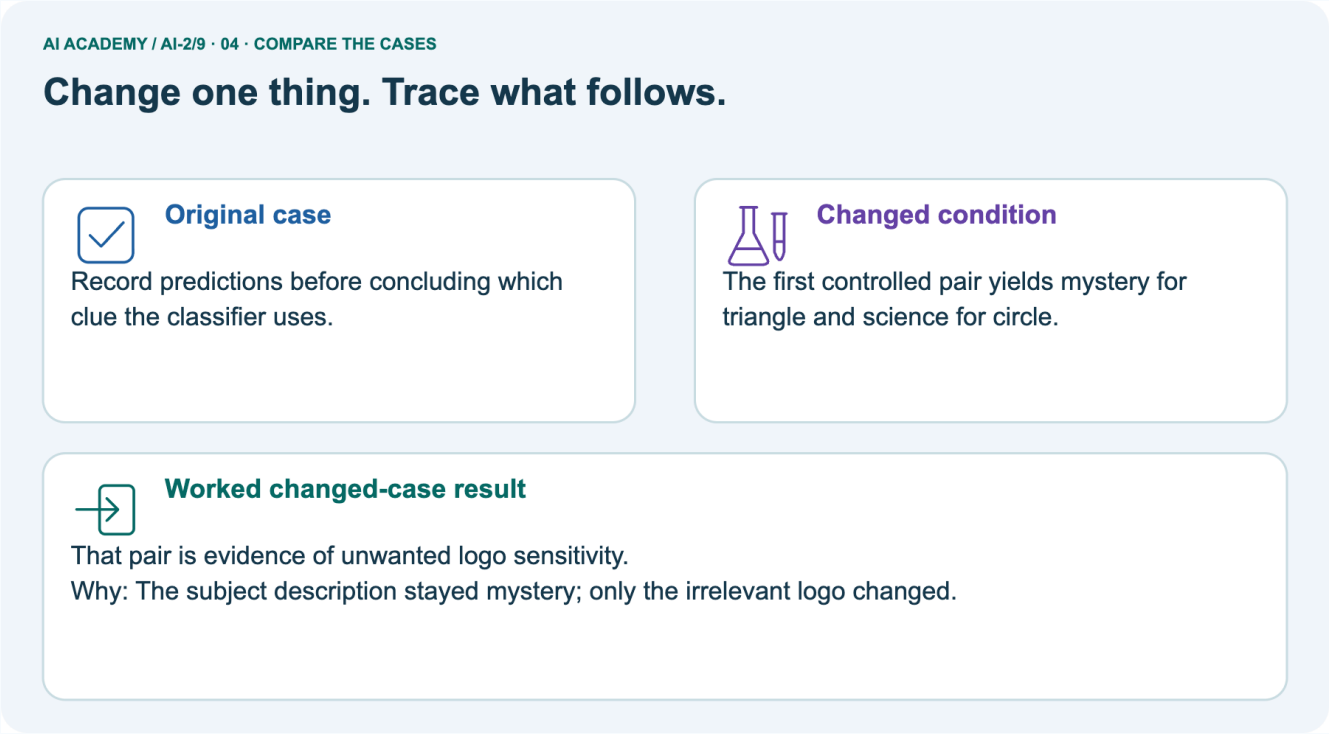
Follow the labelled picture with the matching explanation. Open the vector diagram

The same cards under two **feature** choices. A is 10 by 7 cm and fits | Keep length only | 10 cm. B is 10 by 9 cm and is too large | Keep length only | 10 cm. A is 10 by 7 cm and fits | Keep length and width | 10 cm and 7 cm.

B is 10 by 9 cm and is too large | Keep length and width | 10 cm and 9 cm

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



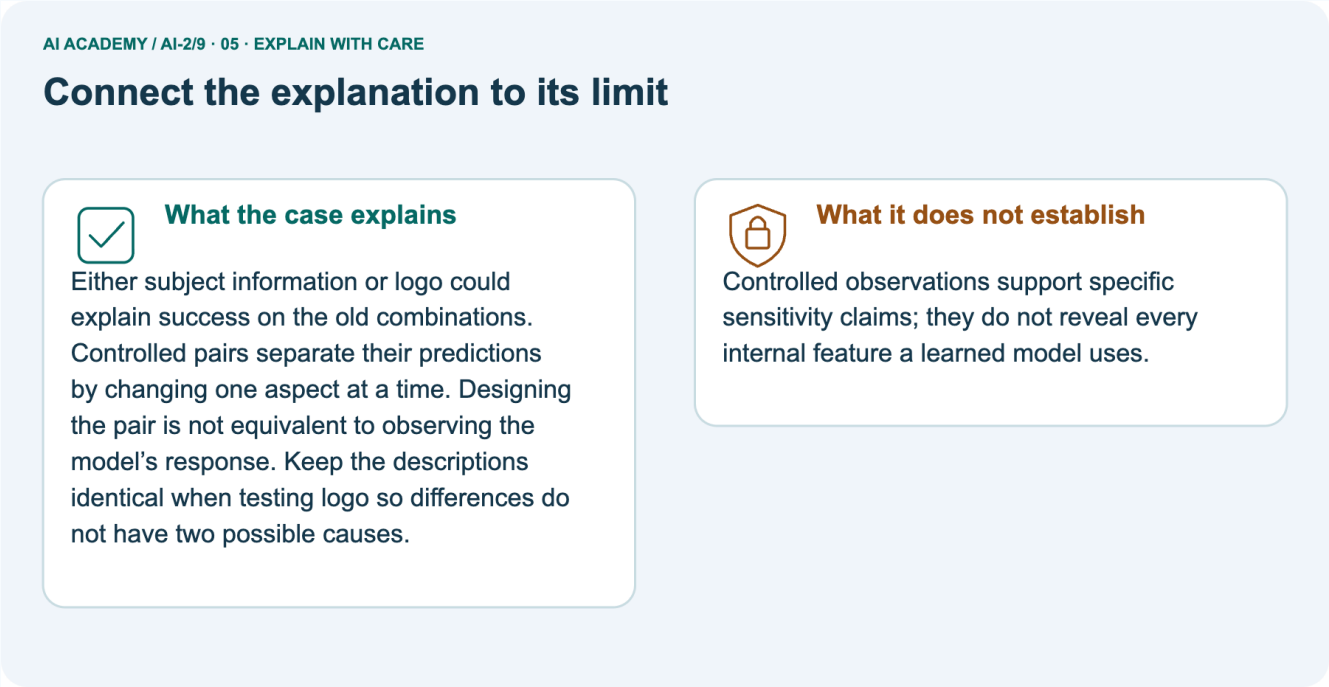
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Record **prediction**s before concluding which clue the classifier uses. Changed condition: The first controlled pair yields mystery for triangle and science for circle.

Expected result: That pair is **evidence** of unwanted logo sensitivity. Reason: The subject description stayed mystery; only the ir**relevant** logo changed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

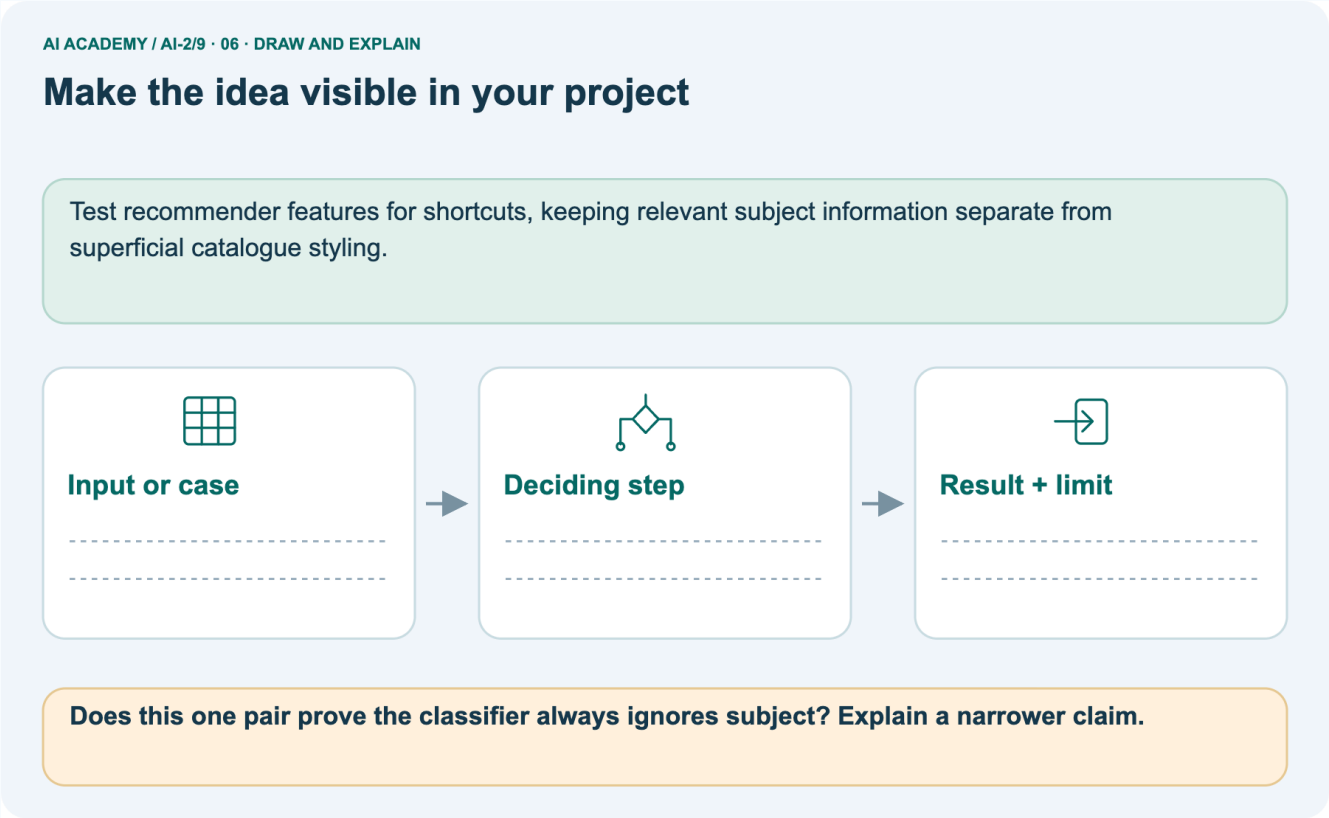
Either subject information or logo could explain success on the old combinations. Controlled pairs separate their **prediction**s by changing one aspect at a time.

Designing the pair is not equivalent to observing the model's response. Keep the descriptions identical when testing logo so differences do not have two possible causes.

Limit: Controlled **observation**s support specific sensitivity claims; they do not reveal every internal **feature** a **learned model** uses.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Test recommender **feature**s for **shortcut**s, keeping **relevant** subject information separate from superficial catalogue styling. Draw your **input**, deciding step, result and limit.

Supported question: Does this one pair prove the classifier always ignores subject? Explain a narrower claim.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Features and Distractors: the case

02. Follow the deciding steps

03. The same cards under two feature choices

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.